

Name: _____
Surname: _____
Date: _____

Intructions

Always assume each code is run independently, so variables do not persist between exercises.

Questions 1-6: Multiple Choice (5 points each)

Circle the correct answer for each question, as each question has **only one** correct answer. A correct answer awards 5 points. There is no penalty for wrong choices.

Questions 7-9: Debugging (10 points each)

These exercises show code cells with one or more errors. You have to find these errors, explain why they happen, and suggest a fix to them.

Questions 10-11: Open Problems (20 points each)

In these exercises, you will encounter problems that are more open-ended and may not have a single correct answer. Your task is to explain how you would approach each problem by breaking it down into smaller, manageable steps.

Describe how you would structure your code, which logical steps are needed, and what kind of Python tools or constructs (e.g., loops, conditionals, functions, lists, etc.) you would use. You are not required to write full code, but doing so is encouraged if it helps clarify your ideas.

1. Exercise 1.
2. Exercise 2.

3. Exercise 3

4. What will be the output of the following code?

```
1 my_list = [0, 1, 2, 3, 4, 5]
2 my_list += [6, 7, 8, 9]
3 print(my_list[4:])
```

- (a) [4, 5]
- (b) [4, 5, 6, 7, 8, 9]
- (c) [4, 5, [6, 7, 8, 9]]
- (d) Error.

5. Which of the following Python structures is not mutable?

- (a) Lists.
- (b) Tuples.
- (c) Sets.
- (d) Dictionaries.

6. Which one of the following statements about dictionaries is **FALSE**?

- (a) We can initialize an empty dictionary using curly brackets {}.
- (b) Keys in a dictionary must be unique.
- (c) Dictionary keys can only be strings.
- (d) A dictionary can contain other dictionaries inside.

7. Exercise 7

8. Exercise 8

9. Exercise 9

10. Exercise 10

Continue exercise 10

11. Exercise 11

Continue exercise 11